

Answers:

1. What is a matrix and how does it differ from a vector?
 - a. A vector is just a single row or column of numbers, representing the distance along each axis. A matrix, on the other hand, I like to think of as an array of vectors, and can represent transformations between spaces.
2. What does it mean that matrices represent linear transformations?
 - a. Generally it means that we can use matrices to map points from one space to another, while maintaining certain geometric properties (such as lines remaining parallel to each other after the transformation).
3. What is an identity matrix, and what does it represent in terms of linear transformations?
 - a. It's a square matrix with 1's on the diagonal, and 0's everywhere else. As a transformation, it represents a sort of "do nothing" transformation.
4. If I have a matrix $M = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ and I transform a point P by M, what effect will this have? Hint: Just look at the columns of the matrix as basis vectors.
 - a. This will be a 90° rotation
5. If you have a matrix $M = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$, and we transform a point P by M, what effect will this have?
 - a. 180° rotation
6. If I wanted to mirror around the Y axis, ie. (3, 2) becomes (-3, 2), and (-4, -2) becomes (4, -2), how would you construct that matrix?
 - a. $\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$
7. Construct a 2x2 matrix that rotates 60°
 - a. $\begin{pmatrix} 0.5 & -0.866 \\ 0.866 & 0.5 \end{pmatrix}$
8. Construct a 2x2 matrix that stretches along the x axis by 3.
 - a. $\begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}$
9. Construct a general 2x2 matrix that rotates by θ , where θ is assumed to be in radians
 - a. If we just go with the basic idea that the first column is the x axis, and the 2nd column is the y axis, then a point defined by θ is simply $(\cos(\theta), \sin(\theta))$, and the y-axis is $\pi/2$ radians rotated from θ , so that would be $(\cos(\theta + \pi/2), \sin(\theta + \pi/2))$

So the matrix would look like this: $\begin{pmatrix} \cos(\theta) & \cos(\theta + \frac{\pi}{2}) \\ \sin(\theta) & \sin(\theta + \frac{\pi}{2}) \end{pmatrix}$

There IS a general version of a rotation matrix that looks like this:

$\begin{pmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{pmatrix}$ which we'll cover in a later chapter. The reason it looks different is that it's just using some properties of sin/cos to reduce the number of calculations. Mathematically, it's identical to what we have.

10. What is a shear transform?

- a. It's a linear transform that creates a skewed or slanted effect. More precisely, it displaces points in a direction parallel to the shear direction, and it's applied proportionally to the direction of the shear. I.e. If you're shearing along x, points in y won't be affected.

11. Given 2 matrices A, B, and the result $AB=C$, where C is the result of multiplying matrices A & B together. In what order are the linear transforms applied? A, then B, or B then A?

- a. B, then A.

12. Given T, R, S, and the result $TRS = C$, which is the result of multiplying those 3 matrices together. In what order are the linear transformations T, R, and S applied?

- a. S, then R, then T

13. Given matrices $A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, without multiplying this out, just visualizing the basis vectors and applying them, what will the final result be?

- a. The result will be the identity matrix. A is a 90° rotation, while B is a -90° rotation, cancelling each other out.